

STEP 12 - BUSINESS ANALYSIS & INSIGHTS

12.1 - COVID Recovery Analysis

Question

How severely did COVID-19 affect passenger traffic, and which regions recovered fastest?

Answer

COVID-19 caused an unprecedented collapse in global passenger demand in 2020.

The strongest impact was in **Asia Pacific**, where RPK fell from approximately **64 billion in December 2019 to 9 billion in April 2020**, representing roughly an **86% decline**.

The recovery was highly asymmetric:

Region	Approx. Recovery to 2019 Baseline
Middle East	2022
North America	2023
Europe	2024
Asia Pacific	2024–2025
Latin America	2024–2025
Africa	2024–2025

Business Insight

The COVID-19 shock affected all aviation markets, but recovery speed differed significantly by region. Middle Eastern aviation recovered earliest, while Asia Pacific experienced a prolonged recovery because of extended travel restrictions and China's COVID policies.

12.2 - Airline Revenue Performance

Question

Which airlines demonstrated the strongest revenue performance?

Answer

Major global airlines such as **Emirates, Delta, United, American Airlines, Lufthansa Group, IAG and Qatar Airways** represent significant revenue-generating businesses in the dataset.

The most important pattern is not simply the absolute revenue ranking but the **COVID shock and subsequent recovery**.

For example:

- Delta: approximately **\$47B in 2019 → \$17B in 2020**

- Lufthansa Group: approximately **\$40B → \$15B**
- Cathay Pacific experienced one of the strongest disruptions.

Business Insight

Scale did not protect airlines from COVID. Large network carriers experienced dramatic revenue declines because their business models depend heavily on international and long-haul traffic.

12.3 - Legacy vs Low-Cost Airlines

Question

Did COVID change the structural margin gap between legacy and low-cost airlines?

Answer

Before COVID, low-cost carriers generally operated with stronger structural margins because of:

- Lower operating costs
- Higher aircraft utilization
- Simplified fleets
- Point-to-point networks
- Lower service complexity

COVID significantly disrupted this advantage.

During 2020–2021, **both legacy and low-cost carriers experienced severe margin deterioration.**

Business Insight

COVID temporarily compressed the traditional margin advantage of low-cost airlines. However, the underlying cost advantages of the low-cost model remained important during the recovery period.

12.4 - Boeing vs Airbus

Question

How did Boeing perform against Airbus during the 737 MAX crisis and recovery?

Answer

Boeing experienced a dramatic decline in net orders:

Year Boeing 737 Family Net Orders

2013 +1,208

2018 +675

Year Boeing 737 Family Net Orders

2019 -87

2020 -1,026

2024 +569

Meanwhile, Airbus recorded approximately **1,456 A320-family net orders in 2024**.

Therefore, in 2024:

Airbus narrowbody orders were approximately 2.5× Boeing's.

Business Insight

The 737 MAX crisis significantly weakened Boeing's narrowbody order momentum. Airbus strengthened its competitive position, particularly in the narrowbody market, although Boeing continued to maintain a major global installed base and backlog.

12.5 - COMAC C919 Growth

Question

Is COMAC becoming a threat to the Boeing-Airbus duopoly?

Answer

COMAC's C919 trajectory shows significant growth:

Year Orders Deliveries

2018 24 0

2023 250 4

2024 220 13

The strongest market is currently **China's domestic market**.

Approximately **80% of domestic Chinese orders** are associated with the Chinese market according to the dataset methodology.

Business Insight

COMAC represents a growing strategic competitor in the narrowbody market, but it is not yet a full global replacement for Boeing and Airbus. Certification, international support infrastructure, production scale and export penetration remain major challenges.

12.6 - Middle East Airline Growth

Question

How strong is the Middle Eastern aviation growth story?

Answer

The Middle East has been one of the strongest aviation recovery regions.

Major players include:

- Emirates
- Qatar Airways
- Etihad
- Saudia
- Riyadh Air

The dataset shows strong long-term growth for Qatar Airways:

\$7B in 2010 → approximately \$22B in 2024

Emirates generated approximately **\$33B revenue in 2024** according to the project dataset.

Business Insight

Middle Eastern airlines are increasingly competing through global hub connectivity rather than only domestic demand. Airport infrastructure, fleet expansion and government-backed aviation investment are supporting the region's ambition to become a global aviation hub.

12.7 - Route Economics

Question

Which routes should airlines prioritize?

The basic revenue model is:

Route Revenue

=

Passengers × Average Fare

Your dataset also provides:

- Distance
- Frequency
- Passengers
- Average fare
- Revenue
- Main airlines

Three useful route categories

Gold Route

High passengers

+

High average fare

=

High revenue potential

Volume Route

Very high passengers

+

Lower fare

=

Volume-driven economics

Premium Route

Lower passengers

+

High fare

=

Premium economics

Business Insight

Airlines should not evaluate routes solely by passenger volume. A route with fewer passengers can generate greater revenue if average fares are significantly higher. Route profitability analysis should additionally incorporate fuel, aircraft utilization, airport charges and operating costs.

12.8 - Regional Market Analysis

Question

Which regions represent the strongest aviation growth opportunities?

Based on the project's regional traffic trajectory:

Middle East

Strongest early recovery and hub expansion.

North America

Rapid post-COVID recovery.

Europe

Strong recovery but more gradual than North America.

Asia Pacific

Huge long-term demand but slower COVID recovery.

Latin America & Africa

Potential growth markets with different levels of infrastructure and connectivity.

Business Insight

Asia Pacific represents one of the largest long-term demand opportunities because of its population and growing air travel market, while the Middle East offers strong hub-driven growth and international connectivity opportunities.

12.9 - Aviation Incident Analysis

Question

Has commercial aviation become safer?

Your dataset contains **40 selected major incidents**, so we should be careful here.

We cannot conclude global aviation safety from only 40 incidents.

However, we can analyze:

- Incident frequency within the dataset
- Fatalities
- Aircraft manufacturers
- Incident severity
- Geopolitical events
- Yearly patterns

Important incidents include:

- MH17
- PS752
- FR4978
- AS1282
- JL516
- AA5342

- DL4819
- AI171

Business Insight

The selected incidents demonstrate that aviation risk is not limited to technical failures. Geopolitical conflict, airport operations, aircraft design and regulatory oversight can all contribute to major aviation events.

12.10 - Final Executive Recommendations

This is the part you can put directly into your **Executive Summary**.

Recommendation 1 - Monitor Boeing's recovery

Boeing should be monitored through:

- Net orders
 - Deliveries
 - Backlog
 - Production rate
 - Narrowbody market share
-

Recommendation 2 - Invest selectively in high-growth regions

The strongest opportunities are:

Asia Pacific + Middle East

because of long-term passenger demand and international connectivity.

Recommendation 3 - Don't judge routes only by passengers

Use:

Passengers

+

Fare

+

Revenue

+

Distance

+

Frequency

+

Operating Cost

to determine route attractiveness.

Recommendation 4 - Track low-cost airline economics

Low-cost carriers should be evaluated using:

- Operating margin
 - Load factor
 - Fleet utilization
 - Passenger growth
 - Revenue per passenger
-

Recommendation 5 - Monitor COMAC

COMAC should be treated as an **emerging strategic competitor**, especially in China, rather than an immediate global replacement for Boeing and Airbus.

Final Executive Summary

The analysis demonstrates that the global aviation industry experienced three major structural disruptions: the COVID-19 demand collapse, the Boeing 737 MAX crisis, and subsequent supply-chain and geopolitical challenges. COVID-19 created significant regional differences in recovery, with the Middle East recovering earlier while Asia Pacific experienced a prolonged recovery. Airbus strengthened its narrowbody competitive position during Boeing's MAX-related disruption, while COMAC's C919 emerged as a growing competitor in China's domestic market.

The analysis also shows that airline performance depends on more than revenue scale. Business model, load factor, passenger volume, fleet size, route economics and regional exposure all influence financial performance. For investors and aviation companies, monitoring aircraft orders, backlog, passenger recovery, route economics and regional growth provides a more complete view of the industry's future.

Global Aviation Industry — 10 Business Questions & Answers

1. How did COVID-19 affect airline revenue and passenger demand globally?

Answer:

COVID-19 caused the most severe disruption in the aviation industry during 2020.

Airline revenues dropped sharply because international travel restrictions, lockdowns, and reduced consumer demand caused passenger volumes to collapse. Passenger traffic also declined dramatically, with **Asia-Pacific particularly affected**, where RPK fell from approximately **64 billion in December 2019 to 9 billion in April 2020**, an approximately **86% decline**.

Airline operating margins also moved deeply negative during 2020.

Business insight:

COVID-19 demonstrated that airline revenue is highly dependent on passenger demand and network utilization. Airlines with stronger liquidity, diversified networks, and domestic markets generally recovered faster.

2. Which regions recovered fastest after the COVID-19 disruption?

Answer:

Recovery was uneven across regions.

Based on the dataset:

Region	Approximate recovery
Middle East	Fastest — exceeded 2019 levels by 2022
North America	Reached 2019 baseline around 2023
Europe	Around 2024
Asia-Pacific	Around 2024–2025
Latin America	Later recovery
Africa	Recovery varied

The **Middle East recovered particularly quickly**, supported by strong international connectivity and major hub airlines.

Asia-Pacific recovered more slowly because prolonged restrictions, particularly in China, continued to suppress international travel.

Business insight:

COVID recovery was not uniform. Government restrictions, domestic demand, international connectivity, and hub strategies played major roles in determining recovery speed.

3. Which airlines demonstrated the strongest financial recovery?

Answer:

The strongest recovery should be evaluated using multiple indicators rather than revenue alone:

- Revenue recovery

- Operating margin recovery
- Passenger recovery
- Load factor
- Fleet utilization

Large international airlines such as **Emirates, Qatar Airways, Delta, and other major network carriers** demonstrated strong post-pandemic recovery in the dataset.

For example, Qatar Airways increased revenue substantially from approximately **\$7 billion in 2010 to \$22 billion in 2024**.

Emirates also reached approximately **\$33 billion in 2024 revenue** in the dataset.

Business insight:

Airlines with strong international networks, premium traffic, major hubs, and high load factors were better positioned to benefit from the post-COVID recovery.

For your actual presentation, replace "strongest" with the exact ranking from your **Airline Performance** dashboard.

4. Which business model—legacy, low-cost, or hybrid—performed best?

Answer:

The dataset indicates that **low-cost airlines generally have stronger structural operating margins during normal operating periods**, primarily because of:

- Lower operating costs
- Higher aircraft utilization
- Simplified fleets
- Additional ancillary revenue
- High seat density

However, COVID-19 affected all business models severely.

Legacy airlines were particularly exposed to the collapse in international and business travel, while low-cost airlines benefited from the faster recovery of domestic and short-haul travel.

Business insight:

There isn't one permanent winner across every period. The performance advantage depends on the market environment.

A useful conclusion is:

Low-cost carriers generally have a structural cost advantage, while legacy carriers benefit from network scale, premium passengers, and international connectivity.

5. How did Boeing's 737 MAX crisis affect its aircraft orders compared with Airbus?

Answer:

The 737 MAX crisis had a major impact on Boeing's order performance.

According to the dataset:

Year Boeing 737-family net orders

2013 +1,208

2018 +675

2019 -87

2020 -1,026

2024 +569

The deterioration became particularly severe in **2019 and 2020** as the MAX was grounded and cancellations increased.

At the same time, Airbus continued to maintain stronger narrowbody demand.

In 2024, the dataset shows approximately:

- **Boeing: +569**
- **Airbus A320 family: +1,456**

Therefore Airbus had roughly **2.5× Boeing's order volume** in that comparison.

Business insight:

The 737 MAX crisis significantly weakened Boeing's narrowbody order position and allowed Airbus to strengthen its competitive position.

6. How has Boeing's position compared with Airbus in aircraft orders, deliveries, and backlog?

Answer:

Airbus has generally maintained a stronger position in the narrowbody market during the post-MAX period.

Boeing experienced significant disruption because of:

- 737 MAX grounding
- Customer cancellations
- Production restrictions
- Regulatory scrutiny
- Supply-chain challenges

- Quality-control problems

Airbus benefited from strong demand for the A320neo family.

However, Boeing remains a major global manufacturer with a substantial installed customer base and backlog.

Business insight:

Airbus gained competitive momentum, particularly in narrowbody aircraft, while Boeing's recovery has depended heavily on restoring production capacity, customer confidence, and delivery performance.

7. Is COMAC emerging as a meaningful competitor to Boeing and Airbus?

Answer:

Yes, but primarily as a developing regional/domestic competitor rather than an immediate global replacement for Boeing and Airbus.

The dataset shows rapid growth in C919 activity:

Year Orders Deliveries

2018 24 0

2023 250 4

2024 220 13

The C919's strongest opportunity is the **Chinese domestic market**, where approximately 80% of domestic Chinese orders are represented in the dataset.

However, COMAC still faces challenges involving:

- International certification
- Global support infrastructure
- Production scale
- International airline adoption
- Supply-chain independence

Business insight:

COMAC represents a long-term competitive threat to the Boeing-Airbus duopoly, particularly in China, but it is not yet operating at the global scale of Boeing and Airbus.

8. Which global routes have the highest passenger volume and revenue potential?

Answer:

The highest-value routes tend to combine:

- Large passenger volumes
- High flight frequency
- Strong business/tourism demand
- High average fares
- Major connecting hubs

The **Top 10 Routes by Passengers** and **Top 10 Routes by Revenue** visuals on Page 4 can identify these routes.

The most attractive routes are not necessarily those with the highest passenger volume.

For example:

A route with very high passenger volume but very low fares may generate less revenue than a lower-volume route with significantly higher fares.

Therefore, the **Fare vs Passenger Volume** scatter plot is important for identifying potentially attractive routes.

Business insight:

Route attractiveness should be evaluated using both volume and revenue, rather than passenger count alone.

9. How has the frequency and severity of major aviation incidents changed over time?

Answer:

The dataset contains **40 selected major incidents**, so it should not be interpreted as a complete measure of global aviation safety.

The dashboard shows incident frequency and fatalities by year and severity.

Major events such as:

- MH17
- COVID-era geopolitical incidents
- 2024 Alaska Airlines 737 MAX 9 door-plug incident
- 2024 JAL Haneda collision
- 2025 DCA mid-air collision
- 2025 Air India crash

can create significant spikes in the dataset.

However, incident counts should also be considered relative to the enormous growth in global air traffic.

Business insight:

The number of selected incidents alone cannot determine whether aviation became safer or less safe. A proper safety analysis should normalize incidents and fatalities against passenger traffic or flight activity.

This is an excellent point to mention in an interview because it demonstrates **analytical thinking rather than simply reading a chart**.

10. What are the major strategic trends shaping global aviation from 2010–2026?**Answer:**

The project identifies several major strategic trends.

1. COVID-19 disruption

The pandemic created an unprecedented collapse in passenger demand and airline revenue.

2. Uneven recovery

Different regions recovered at different speeds because of differences in travel restrictions, domestic markets, and international connectivity.

3. Airbus competitive gains

Airbus strengthened its position in the narrowbody market while Boeing faced the consequences of the MAX crisis.

4. Boeing recovery challenge

Boeing's recovery depends on:

- Production normalization
- Delivery performance
- Quality control
- Regulatory confidence
- Customer confidence

5. COMAC emergence

COMAC's C919 represents the development of a third major narrowbody manufacturer, especially within China.

6. Middle East hub expansion

Airlines such as Emirates and Qatar Airways continue to benefit from strategic geographic positioning and international hub connectivity.

Saudi Arabia is also investing heavily in aviation infrastructure and new airline capacity.

7. Increasing importance of route economics

Airlines need to balance:

Passenger volume + fares + capacity + utilization + operating costs

rather than simply maximizing passenger numbers.

8. Aviation safety and geopolitical risk

The industry is affected not only by operational safety but also by:

- Geopolitical conflicts
 - Airspace restrictions
 - Sanctions
 - Regulatory actions
 - Supply-chain disruptions
-

"Tell me about your Global Aviation project."

answer:

"I built an end-to-end Global Aviation Industry Analysis project covering 2010 to 2026 using PostgreSQL, Excel, Power Query, and Power BI. I analyzed airline financial performance, passenger traffic, aircraft manufacturer orders, global routes, and major aviation incidents.

The main business objective was to understand how major events such as COVID-19, the Boeing 737 MAX crisis, and the post-pandemic recovery affected the aviation industry.

I created a Power BI data model with a Dim Date table and five interactive dashboard pages covering Executive Overview, Airline Performance, Boeing vs Airbus, Passenger & Route Analysis, and Aviation Safety.

One of the key insights was that COVID-19 created a severe and uneven decline in passenger traffic, with Asia-Pacific experiencing one of the largest shocks and the Middle East recovering comparatively quickly. I also found that Boeing's order performance deteriorated significantly during the 737 MAX crisis while Airbus strengthened its narrowbody position. Finally, I analyzed COMAC's C919 as an emerging competitor and evaluated route economics using passenger volume, fares, and revenue."

Step 12 - Business Analysis & Insights - Documentations

Project: Global Aviation Industry Analysis 2010–2026

12.1 Objective

The objective of Step 12 was to convert the Power BI dashboard results into **business insights and actionable recommendations**.

The analysis focused on:

- COVID-19 impact and recovery
- Airline financial performance

- Legacy vs. low-cost business models
- Boeing vs. Airbus competition
- COMAC C919 growth
- Middle East aviation expansion
- Route economics
- Regional passenger demand
- Aviation incidents and safety

The analysis was performed using the Power BI semantic model, DAX measures, interactive filters, and dashboard visuals. A star-schema approach is appropriate for analytical reporting because dimensions provide filtering/grouping context while fact tables provide measurable values.

12.2 COVID-19 Recovery Analysis

Business Question

How severely did COVID-19 affect global passenger traffic, and which regions recovered fastest?

Analysis Approach

The **Passenger & Route Analysis** page was used to compare regional RPK before, during, and after COVID-19.

The analysis compared:

2019 → Pre-COVID baseline

2020 → COVID impact

2021–2026 → Recovery period

Key Finding

Asia Pacific experienced the most severe disruption in the dataset.

The modeled data shows:

December 2019 RPK ≈ 64 billion

April 2020 RPK ≈ 9 billion

This represents approximately an **86% decline**.

Recovery Pattern

Region	Approximate Recovery
Middle East	2022
North America	2023

Region	Approximate Recovery
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Europe	2024
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Asia Pacific	2024–2025
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Latin America	2024–2025
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Africa	2024–2025
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Business Insight

COVID-19 created a severe but uneven shock across the global aviation industry. The Middle East recovered earlier, while Asia Pacific experienced a longer recovery period because of extended travel restrictions and delayed reopening.

12.3 Airline Revenue Performance

Business Question

Which airlines demonstrated strong revenue performance and how severely did COVID-19 affect airline revenues?

Analysis

The Airline Performance page was used to compare:

- Total Revenue
- Passengers
- Load Factor
- Operating Margin
- Fleet Size
- Revenue Growth

Key Findings

The dataset shows significant revenue declines during 2020.

Examples:

Airline	2019 Revenue	2020 Revenue	Approx. Change
Delta	\$47B	\$17B	-64%
Lufthansa Group	\$40B	\$15B	-63%
Cathay Pacific	—	—	Severe disruption

Business Insight

The COVID-19 crisis affected both large and small airlines, demonstrating that airline scale alone does not provide protection against major demand shocks. International and long-haul carriers were particularly exposed to travel restrictions.

12.4 Legacy vs Low-Cost Airlines

Business Question

Did COVID-19 change the structural margin difference between legacy and low-cost airlines?

Analysis

Airlines were compared according to:

- Business model
- Operating margin
- Revenue
- Passengers
- Load factor

Business Insight

Before COVID-19, low-cost carriers generally benefited from:

- Lower operating costs
- High aircraft utilization
- Simplified operations
- Point-to-point networks

During COVID-19, however, both legacy and low-cost airlines experienced significant margin pressure.

Conclusion

COVID-19 temporarily reduced the traditional margin advantage of low-cost airlines, but their structural cost advantages remained important during the recovery period.

12.5 Boeing vs Airbus Analysis

Business Question

How did Boeing's 737 MAX crisis affect its competitive position against Airbus?

Analysis

The Boeing vs Airbus dashboard compared:

- Orders
- Deliveries

- Backlog
- Aircraft families
- Narrowbody/widebody categories

Boeing Order Trend

Year Boeing 737 Family Net Orders

2013 +1,208

2018 +675

2019 -87

2020 -1,026

2024 +569

Airbus recorded approximately **1,456 A320-family net orders in 2024** in the project dataset.

Business Insight

Boeing's 737 MAX crisis caused a major deterioration in narrowbody order momentum, particularly during 2019–2020. Airbus strengthened its competitive position during this period, although Boeing remained a major global aircraft manufacturer with a substantial installed base and backlog.

12.6 COMAC C919 Analysis

Business Question

Is COMAC becoming a serious competitor to Boeing and Airbus?

Analysis

The C919 trajectory was analyzed using orders and deliveries.

Year Orders Deliveries

2018 24 0

2023 250 4

2024 220 13

Business Insight

COMAC is emerging as an important narrowbody competitor, particularly within China's domestic aviation market. However, its global competitive impact remains constrained by international certification, production scale, support infrastructure, and export penetration.

Conclusion

COMAC should currently be considered:

Emerging competitor → not yet global Boeing/Airbus replacement.

12.7 Middle East Aviation Analysis

Business Question

How strong is the Middle East's position in the global aviation industry?

Airlines Analyzed

- Emirates
- Qatar Airways
- Etihad
- Saudia
- Riyadh Air

Key Finding

Qatar Airways increased from approximately:

\$7B revenue in 2010

↓

\$22B revenue in 2024

Emirates reached approximately **\$33B revenue in 2024** in the project dataset.

Business Insight

Middle Eastern airlines are competing through global hub connectivity, fleet expansion, airport infrastructure, and international transfer traffic. The region has therefore developed into one of the most strategically important aviation markets.

12.8 Route Economics Analysis

Business Question

Which routes have the strongest commercial potential?

Key Variables

The analysis considered:

- Annual passengers
- Average fare
- Route revenue
- Distance

- Weekly frequency
- Main airlines

Basic Revenue Formula

Route Revenue

=

Annual Passengers × Average Fare

Route Classification

Category	Characteristics
Gold Route	High passengers + high fare
Volume Route	High passengers + lower fare
Premium Route	Lower passengers + high fare
Weak Route	Low passengers + low fare

Business Insight

Passenger volume alone should not determine route investment. Airlines should combine passenger demand, fare levels, frequency, distance, operating costs and aircraft utilization when evaluating route profitability.

Important Limitation

Your current dataset provides **estimated route revenue**, but it does not contain complete fuel, crew, airport, maintenance and aircraft operating-cost data.

Therefore, this analysis identifies **revenue attractiveness**, not true route profitability.

12.9 Regional Aviation Analysis

Business Question

Which regions provide the strongest future aviation opportunities?

Findings

Middle East

- Strong recovery
- Major international hubs
- Government-backed aviation expansion

North America

- Fast post-COVID recovery

- Large domestic market

Europe

- Strong international connectivity
- Gradual recovery

Asia Pacific

- Very large long-term demand
- Slower COVID recovery
- Significant Chinese domestic market

Business Insight

Asia Pacific represents a major long-term passenger-growth opportunity, while the Middle East represents a strong hub and international-connectivity opportunity.

12.10 Aviation Incident Analysis

Business Question

What does the incident dataset reveal about aviation risk?

The Aviation Safety dashboard analyzed:

- Major incidents
- Fatalities
- Severity
- Aircraft type
- Airline
- Boeing/Airbus involvement
- Geopolitical incidents

Important Events

Examples include:

- MH17
- PS752
- FR4978
- AS1282
- JL516
- AA5342

- DL4819
- AI171

Business Insight

The selected incidents demonstrate that aviation risk can arise from multiple sources, including aircraft technical issues, operational events, geopolitical conflict and airport incidents.

Important Limitation

The dataset contains only **40 selected major incidents**, not every global aviation incident.

Therefore:

The analysis should not be interpreted as a complete measurement of global aviation safety.

12.11 Cross-Industry Business Insights

The combined analysis produces several major conclusions.

Insight 1 - COVID created an asymmetric recovery

Different regions recovered at different speeds.

Business implication:

Airlines should consider regional demand recovery when planning fleet deployment and capacity.

Insight 2 - Boeing's competitive position was affected by the MAX crisis

Boeing's narrowbody order performance deteriorated sharply during 2019–2020.

Business implication:

Aircraft manufacturers need to maintain strong safety, quality and regulatory relationships because reputational events can have long-term commercial consequences.

Insight 3 - Airbus gained narrowbody momentum

The dataset shows stronger Airbus narrowbody order performance during the recovery period.

Business implication:

Airline fleet decisions can create long-term competitive advantages for manufacturers.

Insight 4 - COMAC is an emerging competitor

C919 deliveries increased from essentially zero in the early period to meaningful deliveries by 2023–2024.

Business implication:

Boeing and Airbus should monitor COMAC's production capacity, certification progress and international expansion.

Insight 5 - Middle Eastern hubs are becoming increasingly important

Emirates, Qatar Airways, Etihad, Saudia and Riyadh Air demonstrate the strategic importance of Middle Eastern aviation.

Business implication:

Global airlines need to consider Middle Eastern hub competition when planning long-haul networks.

12.12 Final Recommendations

Based on the analysis, the following recommendations were developed.

Recommendation 1 - Monitor aircraft manufacturer competitiveness

Track:

Orders

Deliveries

Backlog

Narrowbody share

Widebody share

Production rate

Recommendation 2 - Prioritize high-growth regions

Airlines should evaluate capacity expansion in:

Asia Pacific and Middle East

while considering regional regulatory and geopolitical risks.

Recommendation 3 - Evaluate routes using multiple metrics

Don't use passengers alone.

Use:

Passengers

+

Average Fare

+

Revenue

+

Distance

+

Frequency

+

Operating Cost

Recommendation 4 - Monitor airline operating margins

Revenue growth without margin improvement does not necessarily indicate stronger financial performance.

Track:

Revenue

Operating Income

Operating Margin

Passengers

Load Factor

Fleet Size

Recommendation 5 - Monitor COMAC

COMAC should be included in future aircraft-market analysis as its production and certification capabilities develop.

12.13 Project Limitations

The following limitations should be clearly documented.

Financial Data

Some foreign airline financial values are converted to USD and may differ because of exchange-rate effects.

Modeled Data

Several components use interpolation, assumptions and modeled values rather than complete raw historical records.

Route Data

Route passenger and revenue figures are estimates rather than complete OAG schedule-level data.

Incidents

Only 40 major incidents are included.

2026 Data

2026 represents a **partial period**, according to the project description.

No Cargo Airlines

The project primarily focuses on passenger aviation and excludes major cargo carriers such as FedEx and UPS.

No True Route Profitability

Fuel, crew, maintenance, airport and aircraft-specific costs are not fully available, so route analysis focuses primarily on revenue attractiveness.

12.14 Final Project Conclusion

The Global Aviation Industry Analysis 2010–2026 project demonstrates how PostgreSQL, Excel, Power Query and Power BI can be combined to analyze a complex global industry. The analysis identified major structural changes caused by COVID-19, the Boeing 737 MAX crisis, the growth of Middle Eastern aviation hubs and the emergence of COMAC.

The project also demonstrates that aviation performance should be evaluated across multiple dimensions including financial performance, passenger demand, aircraft orders, route economics, regional recovery and major incidents.

The resulting Power BI dashboard provides an interactive analytical framework for executives, investors and aviation analysts to explore industry trends and identify potential business opportunities and risks.

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